

Loose Fruitlet and Fresh Fruit Bunch Detection for Palm Oil Harvest Management

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Abstract— The palm oil industry can be considered one of the crucial industries, especially in Asian countries like Malaysia and Indonesia. The production flow is depending on the efficiency of the palm oil harvest management. The management during harvest time includes collecting and counting the fresh fruit bunch (FFB) and loose fruitlet (LF) production. Thus, in this paper, we proposed a method that able to detect and count the FFB and LF production automatically. The as-proposed method consisted of two parts: (a) fruitlet detection using an image processing prior to harvest and (b) palm oil tree, FFB, and grabber detection using the Faster R-CNN algorithm during harvest time. During the pre-harvest, the number of fruitlets can be determined through the status of the tree either by ready-to-harvest (RTH) or not ready-to-harvest (NRTH). If the status was RTH, the system performed tree detection for tagging purposes. When they started to harvest, the detection system would detect the FFB and grabber. It would then track the grabber until overlaps with the FFB location. Then, the system would add the FFB to the count module. It means the FFB had been taken to the truck. The as-proposed system achieved the detection accuracy of 96.5% for FFB, 99.2% for grabber, and 97.2% for tree.

Keywords—fresh fruit bunch, fruitlet, deep learning, image processing, harvest management

I. INTRODUCTION

Based on the global vegetable oils consumption statistics from 2013 to 2022 [1], palm oil maintains the highest consumed vegetable oil in the world, followed by oil from other plants such as soybean, rapeseed, and sunflower. Palm trees are cultivated in tropical countries such as Indonesia and Malaysia, where these two countries are the largest and second largest producers of palm oil in the world, producing around 85% of global palm oil [2]. Fig. 1 shows the top 10 exporters of palm oil throughout the year 2020 [3]. In Malaysia alone, oil palm plantations have reached an area of 5.85 hectares since it was first planted commercially in 1917 [4]. Surprisingly, this multi-billion-dollar industry still relies heavily on manual labor to perform most of the tasks from planting to harvesting. In fact, Mosquera et al. [5] reported that 90% of the harvesting cost goes to the labor payment. Therefore, when the country faces foreign labor issues nationwide, the industry's productivity is also severely affected. It was reported that the industry may suffer losses of up to USD\$4 billion due to the unpicked fruit rots if the labor issue persists [6]. Therefore, the industry is in desire need to

have automation and mechanization to ease the dependencies on manual workers.

Harvesting is the most labor-intensive task in crop management where it starts with the workers' scout to search for ripe fresh fruit bunch (FFB), cutting, collecting, and gathering the FFB in the field [7-8]. It is critical to start cutting the FFB at the right time to maximize productivity as the oil content of the fruits is dependent on their ripeness or maturity level. Traditionally, the right time is determined by counting the LF on the ground. When there are 1-10 fruits on the ground, it indicates that the bunch is already ripe and needs to be cut immediately. This harvesting task generally will be done by a harvest team or by an individual worker. If the harvest is performed by an individual worker, the judgement of FFB maturity will be based on his experience which over time, he might lose his judgement and consistency in consideration, which translated to low labor productivity [9]. In this study, an automated system for determining the harvest-readiness FFB based on detecting and counting loose fruitlet (LF) on the ground was presented. The system can be used to initially estimate and detect the trees that were ready for harvesting. Thus, the plantation manager can use the information to plan for the right resources to perform the harvest task. The worker can directly go to the identified tree to start cutting the bunch. This will indirectly help to increase labor productivity.

During the cutting activity, it was observed that there would be more LF detached from the bunch as it hits the ground. It was estimated that the majority of fruitlets will be scattered around 1.5 to 2.5m from the impact point [10]. These LF normally were neglected because of their small size as compared to the bunch. However, researches showed that these LF contained most of the oil content as compared to other attached fruitlets, thus they need to be collected. A study [11] showed that the uncollected 20 LF per bunch can contribute between 0.97% to 0.37% of OER reduction dependent on the age of the tree. This is because these LF normally is from the outer layer of the bunch which contributed to 40% of the oil content of the whole bunch as compared to the middle and the inner layers [12]. Uncollected LF also will cause another problem as they will turn into weeds or volunteer oil palm seedlings (VOP) [13], which requires another effort to clean them up in the future. Thus, it is important to collect these LF after the cutting activity to maximize crop productivity. However, this collecting task is tedious and requires a lot of manual work. For that reason, in this study, the as-proposed system is extended to also detect and estimate the density of the LF during the FFB collection

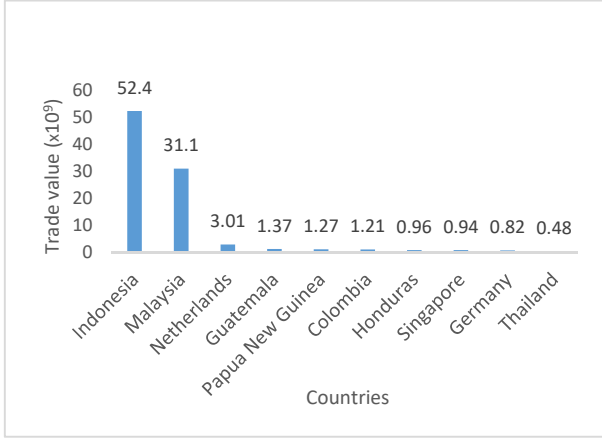


Fig. 1 Top 10 exporters of palm oil in 2020 [3]

activity. The as-gathered density estimation can be used to properly manage the human and machinery resources required to perform the loose fruitlet collection. The information also can be used to estimate the yield productivity as well as to evaluate labor effectiveness.

In summary, this paper is focusing on two-folds; (1) proposing an automated method for detecting and counting LF on the ground to determine the right time to start cutting the FFB, and (2) an automated method to estimate the number of FFB collected for each tree and density of LF on the ground during the FFB collection period. The information can be used to assist in harvest resource planning and management as well as to increase labor productivity during harvesting.

II. METHODOLOGY

This paper proposed a harvest management system that consisted of two parts; (a) method to detect and estimate the number of LF before harvesting by starting to identify tree which is ready for cutting, and (b) method to detect and count the collected bunch for each tree and estimate the density of LF on the ground during the FFB collection activity. The as-proposed process was described in the pseudocode provided in Fig. 2.

The input to the process is the image sequence and the input from the user indicates the current activity either the pre-harvesting or during harvesting time. If the activity is pre-harvesting, the algorithm will first load the tree detection model and detect the oil palm tree. The as-detected tree will be tracked as it appears in the camera view until it leaves the view. As the distance between each tree is quite far (7 to 9 meters apart) and assuming that the scanning path is only in one forward direction, then a simple tree tracking algorithm can be applied. To illustrate the tracking step, refer to Fig. 3. In Fig. 3, the red box indicates the tracking area. Only tree which appears within this tracking area will be tracked and analyzed. As the camera is moving in one direction (e.g., left to right), a new tree will appear at the rightest part of the image first and continue moving to the left part of the image before disappearing. All the frames, while the tree is tracked, will go through the fruitlet detection step. These frames will be further cropped to only include the area within the tree

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Input: image sequence
Output: fruitlet, tree, FFB and grabber
Initialization: fruitlet_count = 0, detected_ffb = 0, collected_ffb = 0
Get user input: h = 0 for pre-harvest or h = 1 for harvesting
Do While there is image to be processed
  Read current image
  If h == 0: //pre-harvest
    Load tree detection model
    Run inference on current image
    If tree is detected:
      Get tree location -> (xmin, ymin, xmax, ymax)
      If tree location is within tracking area:
        Perform tree tracking
        Crop image for fruitlet detection
        Perform fruitlet detection
        Count number of fruitlets -> fruitlet_count
        Store the fruitlet_count for the same tracked tree
      If tracked tree is leaving tracking area:
        Estimate the final fruitlet count ->
        final_fruitlet_count
      If final_fruitlet_count > 3:
        tree_status = RTH
      Else:
        tree_status = RNTH
  Else If h == 1: //harvesting
    Load tree, FFB & grabber model
    Run inference on current image
    If tree is detected:
      Perform fruitlet detection
      Estimate fruitlet density
    If FFB is detected:
      Get FFB locations
      Count FFB -> detected_ffb
    If grabber is detected:
      Get grabber locations
    For each detected FFB:
      Compute overlapping between FFB and grabber boxes ->
      intersection_area
      If intersection_area > 0:
        Increase number of collected FFB -> collected_ffb++

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Fig. 2 Pseudocode of the overall algorithm

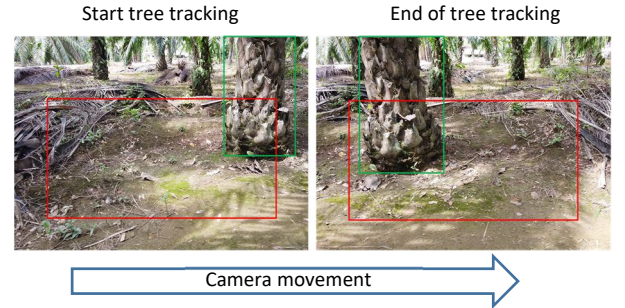


Fig. 3 Illustration of a simple tree tracking algorithm

trunk before passing to fruitlet detection. Fig. 5 and Fig. 6 illustrate the method for detecting LF on the ground by analyzing the color. The method for detecting fruitlets was further discussed in section A. The final estimated number of LF for the tracked tree will be determined based on the most occurrence number of detected fruitlets per frame. Next, if the estimated number of fruitlets is greater than a certain threshold, in our case $t=3$, then the tree will be tagged as ready-to-harvest (RTH). Otherwise, it will be tagged as not-ready-to-harvest (RNTH).

Next, by referring to Fig. 2, if the current activity is for harvesting, similarly, the algorithm will first load the tree detection model as used in the pre-harvest process. The tree detection model is utilizing pretrained faster RCNN model, which has been trained using the large COCO database. Using the transfer learning approach, the model was then retrained



Fig. 4 Sample of three density levels of loose fruitlets (low, medium, high from left to right)

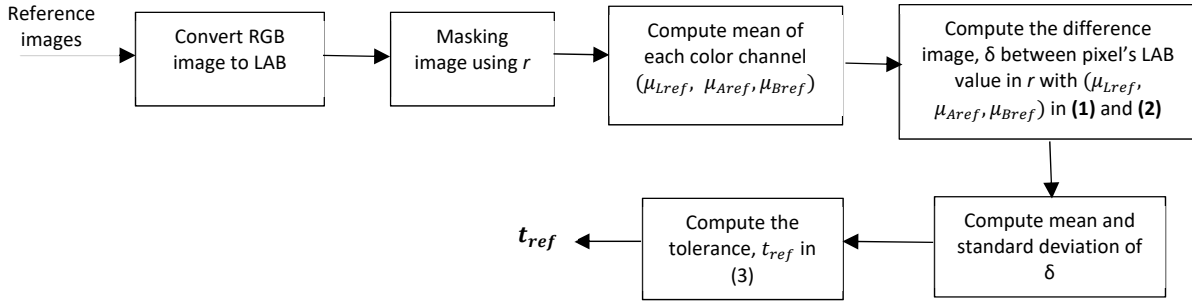


Fig. 5 Offline process to determine fruitlets detection parameters

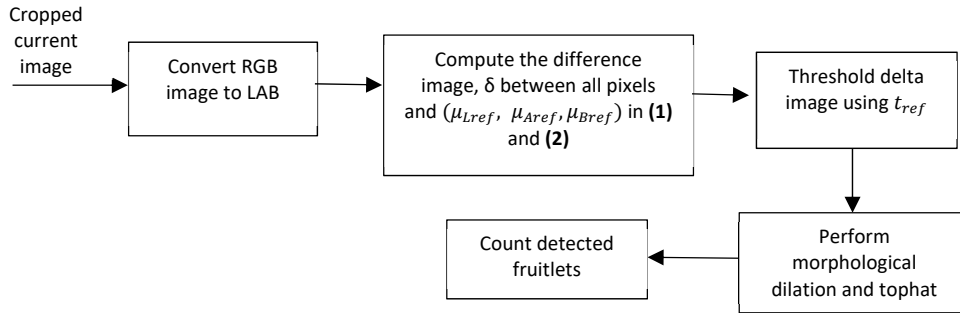


Fig. 6 Online process to detect fruitlets

using our datasets which give one out of three output labels as either oil palm tree, FFB, or grabber. Further discussion on this detection model was provided in section B. Next, if FFB or/and grabber were detected in a frame, the location of these detected objects in the frame will be extracted and checked against each other to find out whether they are overlapped or not. The overlap was determined by finding the pixels areas that were common to both bounding boxes of detected FFB and grabber, indicated by the *intersection area*. If in any frame, the *intersection area* is more than zero, it was assumed that the grabber has successfully grabbed and collected the FFB, thus, the number of collected FFB, *ffb_cnt* will be increased. Then, if there was detected FFB in the frame, the process continues to detect LF on the ground and estimate their density. Three density level was considered in this work: (1) low for the number of fruitlets less than 50; (2) medium for the number of fruitlets between 50 to 100; and (3) high for the number of fruitlets more than 100.

A. LF detection algorithm using image processing approach

The process for detecting LF using the image processing approach was divided into two parts: (1) offline process to determine the detection parameters, which include the reference mean for each Lab color channel (μ_{Lref} , μ_{Aref} , μ_{Bref}) correspond to fruitlets and the reference tolerance value, t_{ref} ; and (2) online fruitlets detection process. Each process was clearly illustrated in Fig. 5 and Fig. 6, respectively.

By referring to Fig. 5, the offline process requires reference images and their corresponding ROI images. ROI images are the binary images to indicate the annotated bounding box of the fruitlets in reference images. The process always starts with the RGB image conversion to Lab color space. In this work, Lab color space was chosen as it shows good recognition. Then, the ROI region in r was used to mask out all pixels other than the pixels within fruitlets bounding boxes in the Lab image. Based on these pixels, the means of each color channel were computed, denoted as μ_{Lref} , μ_{Aref} and

μ_{bref} . Next, the difference between each pixel's value to this reference means was calculated as shown in (1), where n is representing each color channel. The final difference value for each pixel was calculated using (2). The resultant image from this operation was denoted as δ image. Next, the mean, μ_δ and standard deviation, σ_δ of these δ images were computed. These two values were then used to determine the reference tolerance value, t_{ref} that will be used during the online fruitlet detection as the threshold value. The tolerance value was calculated as shown in (3). The lowers the tolerance value, the weaker the detection.

$$d_n(i, j) = |p_n(i, j) - \mu_n|; n = \{L, a, b\} \quad (1)$$

$$d(i, j) = \sqrt{d_l(i, j)^2 + d_a(i, j)^2 + d_b(i, j)^2} \quad (2)$$

$$t_{ref} = \mu_\delta + \sigma_\delta \quad (3)$$

After the detection parameters had been set, the online fruitlet detection can be activated. By referring back to the pseudocode as presented in Fig. 2, the LF detection would only be triggered if an oil palm tree was detected in the current frame. The location of the detected tree can be used to determine the potential area of LF. Thus, the current image will be cropped to this potential area before detecting the LF. This potential area may cover the whole weeded area which was around 1.5 to 2m from the trunk [14]. But for simplicity, in this study, the potential area was covered with the half bottom of the image. This cropped image will be the input to the detection process. Similar steps were done by converting the input image to Lab color space. The difference image, δ was generated by finding the distance between each pixel's Lab value to the reference means ((1) and (2)) followed by thresholding the difference image using the reference tolerance value, t_{ref} . If the pixel value, $\delta(i, j)$ is less than the t_{ref} , then the pixel (i, j) is set to be 255, otherwise, it is set as zero, resulting in a binary image. Morphological operations including the dilation and top-hat were then applied to the binary image to remove the noise. In the final step, the as-connected component was applied to the binary image and the number of connected regions is corresponding to the number of detected fruitlets in the current image.

B. Tree, FFB and grabber detection

In this section, oil palm tree, FFB, and grabber were the object of interest. The method used to perform these object detection tasks was using a Faster-RCNN detection model. The model was pre-trained using the large-scale object detection dataset (MS COCO). The model was further retrained via a transfer learning approach to include the knowledge from our own datasets. The architecture of Faster-RCNN consisted of three main modules:

- 1) *Convolutional layer*: for extracting appropriate features. In this part, filters were trained to extract features in the image. The weights of these filters were determined from the COCO datasets.
- 2) *Region proposal network (RPN)*: for generating region proposals. In the RPN section, a small neural network works on the last feature map of convolutional layers to generate several candidate region proposals. These proposals were not the final region, they would be filtered out using the objectness score. To cater to different object sizes, several

sizes of anchors with different aspect ratios and scales were introduced.

3) *Prediction*: for detecting the object in the proposed regions. The region proposed by RPN could be input to the fully connected layer (FC) to predict the object class. In this work, only these FC layers were unfrozen during the model retraining. The last FC layer was also changed to output only based on the three object categories.

In this study, we collected 5013 images with the size of 700 x 1000 from a smallholder's plantation area near Sepang, Selangor Malaysia. Among the retraining strategies tested were varying distribution of data splitting between training, testing and validation and varying the number of training steps as listed in Table I. Data splitting distribution was written in the form of (%training, %testing, %validation). For example, (80,10,10) which means that the data was split between 80% training, 10% testing, and the last 10% for validation. On the other hand, the training steps had been tuned gradually from 15×10^3 to 30×10^3 and ended with 50×10^3 .

TABLE I. HYPER-PARAMETERS TESTED DURING MODEL RETRAINING

Hyper-parameter	Experiment		
	1	2	3
Data splitting	(80, 10, 10)	(70, 20, 10)	(60, 20, 20)
Steps	15,000	30,000	50,000

III. RESULTS AND DISCUSSION

In the overall system, a user would give input to the algorithm either 0 for pre-harvest or 1 for during harvest. Pre-harvest would determine the status of palm oil trees based on the fruitlets detection output and tree detection for tagging, while during the harvest process, FFB and grabber detection results were evaluated. During pre-harvest, the RTH or NRTN status of the oil palm tree was determined by the number of LF on the ground that was closed to the tree. The LF counting accuracy was evaluated based on the number of correctly detected to the total number of fruitlets on the ground (in (4)). It can be defined as a confusion matrix; the ratio of the sum of true positives (TP) and true negatives (TN) to the total number of trials (sum of TP, false positives (FP), false negatives (FN), and TN). TP represent correctly detected fruitlet (positive class) while TN represents correctly detected background (negative class). FP denotes false detected fruitlets such as dried leaves falsely detected as fruitlet, while FN is miss detection. The example of the calculation is shown in (5) below and by referring to Fig. 7 (a), where there were 18 fruitlets detected (blue boxes), two FN were highlighted in the red circle, and two closely-spaced fruitlets were detected as one object (considered as one TP and one FN):

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \times 100 \quad (4)$$

$$\frac{18+0}{18+0+0+3} \times 100 = 85.7\% \quad (5)$$

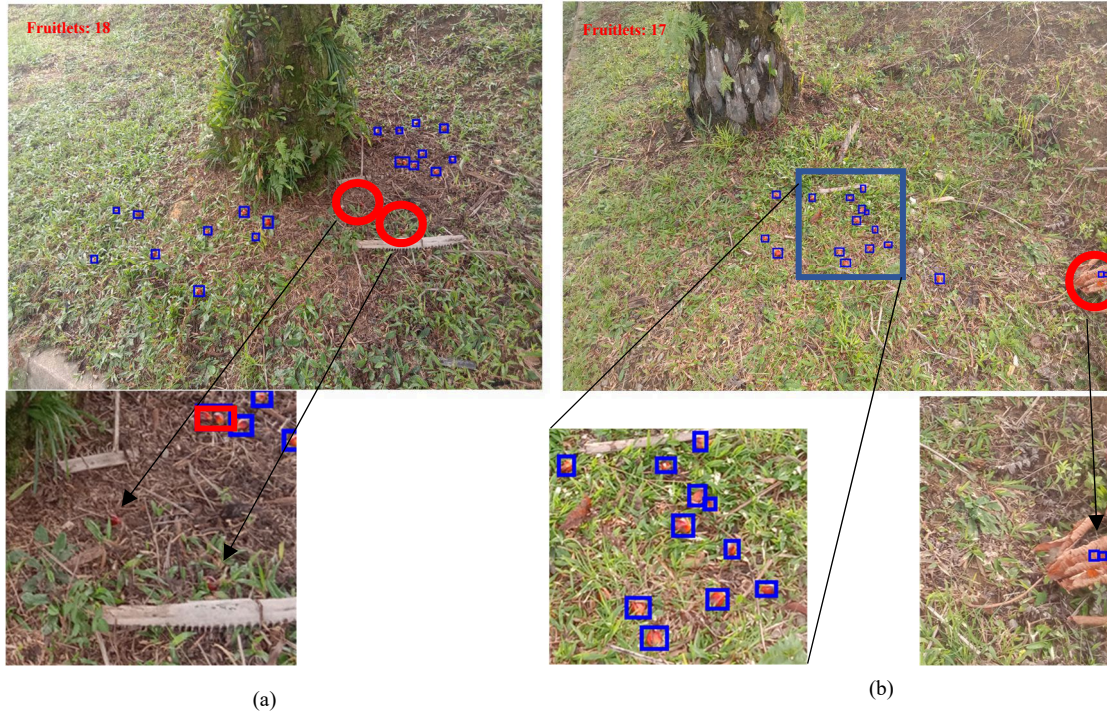


Fig. 7 Illustration of fruitlets detection result; (a) red circles show FN, while red box shows sample of two fruitlets detected as single fruit (one TP, one FN); (b) red circle shows two FPs



Fig. 8 Sample tree, FFB and grabber detection

Another example was given in Fig. 7(b), where there were 17 LF detected with two of them being false detection (FP) as highlighted in the red circle. Thus, the accuracy of this sample image was 88.24%. In total, we have evaluated 10 test images for a total of about 200 LF with an average accuracy achieved was 92.65%. By referring to Fig. 7, most of the LF were successfully detected using the as-proposed method. False detection (FP) was contributed by the background objects with similar colors as fruitlets. For example, in Fig. 7(b), false detection was due to the dried leaves. As the proposed fruitlet detection method was using color analysis for detection, thus, it was expected that any object with an almost similar color range to a fruitlet will be falsely detected as well. However, cropping the input image based on the as-detected tree location will help to reduce the occurrence of false detection. On the other hand, for multiple fruitlets that were closed to each other, they may be detected as single fruitlet as shown in Fig. 7(a). In this case, one fruitlet in the group would be considered as correctly detected while the others were considered FN. This

problem can be further improved by exploring the other features of fruitlet such as the shape feature.

Next, the system is evaluated in terms of the tree, FFB, and grabber detection accuracy. Sample detection results are shown in Fig. 8, where the total bunch detected in the current frame was denoted as the FFB detected, while collected FFB is showing the total number of bunches picked up by the grabber and put into the truck. The latter is based on the overlapping analysis between grabber and FFB bounding boxes. The detection accuracy in terms of the mAP at 50% is shown in Table II, Table III, and Table IV for different data splitting distribution and steps parameters.

TABLE II. DATA SPLITTING WITH (80, 10, 10)

Steps ($\times 10^3$)	mAP		
	FFB	Grabber	Tree
15	95.5	96.8	95.9
30	96.1	97.3	96.7
50	96.6	98.3	97.1

The results show that the best mAP of FFB (96.6%), grabber (99%), and tree (97.1%) was achieved using 70% training data, 20% test data, and 10% evaluation data when steps are at 50,000.

Since the detection method would be based on the color analysis, the detection results will be affected when there were other background objects with a similar color range as the fruitlets such as dried leaves in the image. To reduce the detection error, a cropping strategy is applied, whereby the tree detection box was used to indicate the potential area of the LF. The image would then be cropped to cover the area near the tree trunk before passing to the fruitlet detection step. On the other hand, the as-developed LF detection method can also be applied during the harvesting process to estimate the density of the LF on the ground when the cut FFB was detected. The information can be used to properly manage the resources required to perform the LF collecting activity. Besides that, the system was also coupled with the FFB and grabber detection to estimate the number of as-collected FFB based on the overlapping analysis between the FFB and grabber detected boxes.

TABLE III. DATA SPLITTING WITH (70, 20, 10)

Steps ($\times 10^3$)	mAP		
	FFB	Grabber	Tree
15	95.4	96.8	96.2
30	96.0	98.4	96.7
50	96.6	99.0	97.1

TABLE IV. DATA SPLITTING WITH (60, 20, 20)

Steps ($\times 10^3$)	mAP		
	FFB	Grabber	Tree
15	95.1	96.5	95.8
30	95.9	97.6	96.6
50	96.5	99.2	97.2

IV. CONCLUSION

In this work, the authors have proposed an image-processing-based method for detecting LF on the ground, which can be used to predict the FFB harvest readiness status during the pre-harvesting period. The as-proposed system predicts oil palm tree status either RTH or NRTH according to the number of fruitlets detected. Based on the results presented, the counting accuracy of fruitlet detection achieved 92.65% on average for 10 test images with 200 total fruitlets. The accuracy for FFB, grabber, and tree detection model achieved mAP0.5 of 96.6%, 99%, and 97% using the Faster-RCNN method. Then, the output can be integrated into a virtual map either 2D or 3D map that can show the location and status of the tree to assist and manage the plantation management in terms of labor and cost management.

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